

Design Specification Considerations

Please answer the following questions.

1. What do you want your project to do?
 - *My project will be able to monitor a user's forearm temperature and display the results on a screen in real time. Additionally, if within the available time frame, I will propose and design a self-heating mechanism integrated into the sleeve.*
2. Is the project for you or someone else?
 - *This is a project for the RhythmLink LLC Internship.*
3. If someone else, have you talked to them about design specs?
 - *Not yet.*
4. Are you considering a group project? What is your part?
 - *Yes, this will be a partnership project; I will be working with Kathryn Wu.*
5. Will your project be inside or outside?
 - *Inside*
6. Will your project be portable?
 - *Yes.*
7. Will your project connect to the Internet?
 - *Yes.*
8. Will your project use Bluetooth?
 - *N/A*
9. Does your project use a vinyl cutter?
 - *Yes.*
10. Does your project use a laser cutter?
 - *No.*
11. Does your project use a 3D printer?
 - *Yes.*
12. Does your project use a large CNC machine (Shopbot)?

- *No.*
13. Does your project have intelligence (Arduino, Raspberry Pi, computer)?
- *Yes.*
14. What are your project inputs?
- *PT1000 RTD Temperature sensor*
15. What are your project outputs?
- *1.8" TFT LCD - ST7735*
16. How does your project differ from the project that inspired you?
- *N/A*
17. When was the inspirational project built?
- *N/A*
18. Do you have a tutorial or instructions for your project?
- *Yes, project action items will be provided by RhythmLink*
19. How current is the tutorial?
- *N/A*
20. What is the maximum that you want to spend? No more than \$75.00
- *\$70-100, depending on the cost of textile materials and flex PCB production*
21. What are the dimensions of your project?
- *This is contingent on the average circumference of the user's forearm. As of now, I'm considering a length of 10-12 inches and a width of 8 inches.*
22. What materials will you use?
- *For the physical sleeve:*
 - *Modal and/or fleece for the interior layer (for comfortable skin-to-sleeve contact)*
 - *Nylon for the external sleeve*
 - *For the electronics:*
 - *Kapton flex PCB substrate*
 - *3D-printing filament for electronics casing*
23. Are the parts for your project still available?

- *Yes.*

24. Are the tools you need for the project found in the FabLab?

- *Most tools will be available in the Fab Lab. For the flexible kapton PCBs, I will likely have to order them from a PCB house, such as PCBway.*

25. How will you conceal the electronics?

- *The electronics will be concealed in the fabric. I plan on using flex PCBs and layering fabric around it.*